Assignment #3

Due Mon Feb 23 at 5 PM

Background

- You are asked to:

  Integrate parcel data, FEMA floodplain data, and US Census geographic boundaries to evaluate flood risk exposure of single-family structures and its relationship to parcel improvement value in Brazos County, TX

- You must clearly document:
  1. Your data acquisition strategy
  2. Your data cleaning and preprocessing steps
  3. Your spatial design decisions (projection, geometry transformations, buffering logic, joins)
  4. Your reasoning for analytical choices

- Use the tools we learned so far >> reproducibility and clarity are essential!

Introductions

- This assignment is *intentionally* designed with limited procedural instructions

  One of the primary objectives is for you to demonstrate how you plan, structure, and justify a spatial data engineering workflow

- Grade will consider as much on analytical design decisions as on final outputs
- Same data. Same base assignment. Different focus.
  - DAEN: Build the infrastructure for spatial analytics
  - ISEN: Evaluate the decision of spatial analytics
  - Shared Expectation: Design and justify a defensible spatial workflow
- As you work on the assignment, please be prepared to explain and walk through your decisions if asked!
  - You should be able to clearly describe what you did and why
  - Make sure you truly understand your workflow and results

Part 1: Data Acquisition and Planning

1. Obtain Brazos County Parcel Data https://brazoscad.org/tax-information/gis/
   - Ensure the dataset includes:
     - Parcel geometry
     - Improvement value (or comparable building value metric)
     - Parcel identifier
   - Document:
     - Data source and format
     - Coordinate reference system (CRS)
     - Any missing or inconsistent fields
2. Transform Parcels to Points
   - Convert parcel polygons to point representations
   - Justify:
     - Whether you use centroids, interior points, or another approach
     - The implications of this choice for flood risk classification
     - Potential bias introduced by large or irregular parcels
   - Discuss:
     - What happens if a parcel intersects multiple flood zones?
     - Does a point representation simplify or distort risk classification?

Part 2: Flood Risk Engineering

1. Acquire FEMA flood hazard layer(s) for Brazos County
   https://floodmaps.fema.gov/NFHL/status.shtml
   - Document :
     - Dataset details, CRS, geometry type
     - Flood zone classification fields
2. Reclassify Floodplain Categories
   - **Reclassify flood zones into 100-year, 500-year, or Outside floodplain**
   - Define:
     - Which FEMA zone codes correspond to each category
     - How ambiguous or special zones are handled
     - Provide a reclassification table in your report
3. Assign a flood risk category to each parcel (*spatial join?*)
   - Should parcels be intersected as polygons?
   - Should parcels be converted to points first?
   - Does buffering make sense? If so, what should be buffered? Parcels or floodplain?
   - Why? Explain your approach

Part 3: Census Geography Integration

1. Acquire county boundaries from the U.S. Census Bureau
   (e.g., TIGER/Line or tigris or tidycensus)
   - Check Brazos County geometry
   - Ensure CRS compatibility with other layers
2. Acquire Census "Places" boundaries
   - Document:
     - Definition of "place"
     - How incorporated and census-designated places differ (if relevant)
3. Assign Parcels to Places
   - Using a spatial join:
     - Assign each parcel to a Census Place
     - Then filter to include only Places located in Brazos County
   - Discuss:
     - How you handle parcels that fall outside any Place
     - Whether to rely on intersection or centroid containment

Part 4: Summary Statistics

- After constructing the final parcel dataset with:
  - Flood risk classification + Place assignment + Improvement value
- Produce the following summary statistics:
  - **4.1 Parcel Counts**
    - Total number of single-family parcels in:
      - 100-year floodplain, 500-year floodplain, Outside floodplain
      - In-risk vs out-of-risk totals (define clearly what qualifies as "at risk")
  - **4.2 Improvement Value Comparisons**
    - Compute average improvement value for parcels:
      - Inside flood risk areas and outside flood risk areas
    - Separate averages for 100-year vs 500-year zones
      - Median values and standard deviation
  - **4.3 Possible questions to discuss:**
    - Are differences economically meaningful?
    - Are there extreme values influencing the mean?
    - Would transformation (e.g., log values) be appropriate?
    - Also:
      - Any tradeoffs between computational efficiency and spatial precision.
      - Potential sources of misclassification, or assumptions that could affect conclusions.

Deliverables

Submit:

1. A reproducible script or notebook (R or Python)
2. A short technical report explaining:
   - Data sources
   - Workflow design
   - Key assumptions
   - Justification for spatial decisions
   - Interpretation / summary results (*i.e., the tables below*)

   (4–5 pages recommended)
3. Tables of summary statistics
4. Final engineered dataset (parcels with flood and place attributes)